DUARTE FONSECA

SOFTWARE ENGINEER & GAME DEVELOPER

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EXPERIENCE

Teacher at Mad Fox Labs **August, 25 – September 26**

- Taught extracurricular STEM modules focused on Godot Game Development, Robotics, and Artificial Intelligence.

Lead Unity Developer & Project manager **September 26 - Present**

Hasty Chefs

- P2P Multiplayer, friends-like with driving and cooking mechanics
- Leading a team of programmers and game designers
- Managing the project, making sure deadlines are respected

MAIN PROJECTS

Ego Fractum - VR **January 2026 - July 2026**

Link to learn More about this project

- Developed an animation system that mapped controler keys to a corresponding finger for each virtual hand, bending them depending on the pressure on the button.
- Made a hand pose animation system for the player interactions with in-game objects.
- Implemented game puzzles.
- Implemented a simple state machine enemy.

Nature Strikes Back - Unity **January 2026 - February 2026**

- Optimized the game so that it was suitable to be played on pc handhelds
- QA and bug fixes

Zombie Meltdown - Unity **July 25 - December 25**

Link to learn More about this project

Gameplay Showcase

- Designed a modular weapon system using Scriptable Objects
- Made an in-editor tool to animate the weapons through Scriptable Objects
- Optimized enemy state updates using a centralized update loop, reducing MonoBehaviour overhead and enabling large-scale enemy waves.
- Made a fun and engaging player controller with different states;

OTHER EXPERIENCE

Coordinator & Community Manager **February 25 – Present**

- Organized online events and social gatherings to strengthen community interaction
- Led the community management for a 244-member student club, ensuring engagement and a safe, inclusive environment

ISEL Game Studios **March 2025 - Present**

- Managed staff and overseed event's production and fostered an active student game dev community
- Founded and lead a game development student's club with 100+ members, organizing game jams, talks, and collaborative projects focused in developing games

SKILLS

Unity | C# | Godot | Unreal | C/C++ | Blender | Java

Git Wokflows | OOP | Unit Testing | User Testing | AI Workflows

EDUCATION

M.Sc in Digital Games Development

University of Leiria (ULO, RUN-EU)

2026 – 2028

BSc in Informatics and Multimedia Engineering

Insituto Superior de Engenharia de Lisboa

2023 – 2026

LANGUAGES

Portuguese - Native

English - Fluent

OTHER PROJECTS

Minecraft Clone - Unity **January 2026 - February 2026**

- Built infinite, optimized voxel world generation using FrBM and chunk-based rendering
- Implemented an efficient mesh generation system with face culling and custom UV mapping

Space Game - Java + Processing (No Engine) **December 2024 - January 2025**

Link to learn more about this project

- Engineered a custom physics engine with collision detection, rigid-body dynamics, and particle systems
- Implemented procedural cave generation using cellular automata and custom particle systems

AI Agents with Machine Learning - Unity **January 2026 - April 2026**

- Developing ML-driven AI agents in Unity with Python integration
- Training agent behaviors for autonomous decision-making

Veterinary Clinic Management System **April 2025 - July 2025**

- Developed a full-stack veterinary clinic management system in collaboration with a teammate, simulating real-world operations.
- Designed a normalized MySQL database schema with ER diagrams and implemented backend logic in Java.
- Integrated the system using JDBC and im plemented user authentication and role-based access control.

IMDb Sentiment Analysis Model **July 2025 - December 2025**

Python, Machine Learning

- Developed a machine learning model to classify IMDb reviews as positive or negative using Python.
- Preprocessed and cleaned large text datasets, including tokenization and feature extraction.
- Trained and evaluated the model to optimize accuracy and performance.
- Built a complete data pipeline from data processing to prediction output